Week :9 (26-30 October 2026)

الأسبوع :# من - إلى

Subject/الموضوع: Physics **Unit Title**/عنوان الوحدة: Motion in Two Dimensions - Projectile Motion **Grade/11: الصف**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 الحصة الأولى	Learning Target: I can describe projectile motion as independent horizontal and vertical motion and resolve an initial velocity into components. (DOK 2)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight big ideas and patterns using organisers.	Please check students' Teams for lesson materials and updates.
Lesson 2 الحصة الثانية	Learning Target: I can calculate flight time and horizontal range for a horizontally launched projectile using vertical and horizontal equations. (DOK 3)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight big ideas and patterns using organisers.	Please check students' Teams for lesson materials and updates.
Lesson 3 الحصة الثالثة	Learning Target: I can resolve an angled launch velocity into components and calculate flight time, maximum height and range for a projectile landing at the same height. (DOK 3)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight big ideas and patterns using organisers.	Please check students' Teams for lesson materials and updates.
Lesson 4 الحصة الرابعة	Learning Target: I can interpret the GRASP challenge, identify its audience, role, goal, situation, product and success criteria, and propose a justified projectile-motion investigation plan. (DOK 3–4)	2.1 Clarify vocabulary and symbols using visuals and supports. 3.2 Highlight big ideas and patterns using organisers.	Please check students' Teams for lesson materials and updates.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a calculator, ruler, pencil and Physics notebook. GRASP instructions and submission dates will be shared on Teams.		

الأسبوع# Week # 9

Technology Integration دمج التكنولوجيا	Use a projected trajectory simulation or video to compare launch speed and angle; distribute the GRASP brief and resources via Teams.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: trigonometry, resolving vectors, quadratic relationships and interpreting graphs. Engineering: designing and evaluating launch solutions.
External Assessment Integration دمج التقييمات الخارجية	NGSS-aligned modelling, quantitative reasoning, evidence-based explanation and engineering-design performance tasks.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL: critical thinking, collaboration, communication and self-management. Promote safe participation and constructive feedback.
Standards المعايير	NGSS HS-PS2-1 (supporting motion and forces modelling); HS-ETS1-2 (engineering design: break a complex problem into manageable components). Science and Engineering Practices: Developing and Using Models; Using Mathematics and Computational Thinking. Projectile calculations are supporting skills rather than the full HS-PS2-1 expectation.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	How can horizontal and vertical motion be modelled independently to predict where a projectile will land? How do model assumptions affect real-world design?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1 — MUST:** Identify the two independent components of motion. **SHOULD**: Draw and label velocity components. **COULD:** Explain independence using evidence. **Lesson 2 — MUST:** Identify known variables and equations. **SHOULD:** Calculate flight time and range accurately. **COULD**: Evaluate changes in launch conditions. **Lesson 3 — MUST**: Resolve initial velocity into x and y components. **SHOULD:** Calculate time, height and range. **COULD:** Compare trajectories and justify a conclusion. **Lesson 4 — MUST:** Identify all GRASP elements and deliverables. **SHOULD:** Draft a feasible plan with calculations and evidence. **COULD:** Justify design choices, constraints and improvements.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء — Global Understanding التفاهم العالمي — Arabic Language اللغة العربية
	المجال Area: UAE innovation, responsible engineering, transport and community safety.
	التطبيق Implementation: Explore a hypothetical UAE emergency-supply delivery challenge, modelling a safe projectile trajectory and evaluating assumptions; use this context for the GRASP task launch.